

The computed tomographic “tree-in-bud” pattern: Characterization and comparison with radiographic and clinical findings in 36 cats

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Abstract

In humans, a CT “tree-in-bud” pattern has been described as a characteristic of centrilobular bronchiolar dilation, with bronchiolar plugging by mucus, pus, or fluid. Aims of this retrospective, descriptive, multi-center study were to characterize the CT appearance of a “tree-in-bud” pattern in a group of cats, and compare this pattern with radiographic and clinical findings. Databases from four hospitals were searched during the period of January 2012 to September 2015 and cats with thoracic radiographs, thoracic CT scans and CT reports describing findings consistent with a “tree-in-bud” pattern were included. Images were reviewed by two veterinary radiologists and characteristics were recorded based on consensus. Clinical findings were recorded by one observer from each center. Thirty-six cats met inclusion criteria. Six cats were asymptomatic, 12 were diagnosed with bronchial disease and 23 were suspected to have bronchial disease. Right cranial and right caudal lung lobes were most commonly affected on both imaging modalities. Localization of the “tree-in-bud” pattern was most often peripheral. On radiographs, the CT “tree-in-bud” pattern often appeared as soft-tissue opacity nodules; their number and affected pulmonary segments were often underestimated compared with CT. In conclusion, the “tree-in-bud” pattern should be considered as a differential diagnosis for radiographic soft tissue opaque nodules in feline lungs. Based on lesion localization and presence or suspicion of a concomitant bronchial disease for cats in this sample, authors propose that the CT “tree-in-bud” pattern described in humans is also a characteristic of bronchial or bronchiolar plugging and bronchial disease in cats.

KEYWORDS

bronchi, feline, mucus, nodule, plug

1 | INTRODUCTION

Feline allergic asthma and chronic bronchitis are common inflammatory lower airway diseases primarily affecting the bronchi, with overlapping clinicopathologic features, but different etiologies.^{1–3} Both cause respiratory signs, such as cough or wheeze, and in the case of asthma, may manifest with episodic expiratory distress. Thoracic radiography plays an important role in the investigation of a cat with suspected bronchial disease, in large part to rule out other cardiopulmonary disorders manifesting with similar clinical signs. Several radiographic features consistent with asthma or chronic

bronchitis have been reported including a bronchial and an unstructured interstitial pattern, lung hyperinflation, and hyperlucency, or lobar atelectasis.^{4–9} Hiatal hernias and rib fractures have also rarely been reported as consequences of severe coughing and respiratory distress.¹⁰ Structured pulmonary soft tissue opacities have seldom been described in asthmatic cats and have been attributed to eosinophilic granulomas or mucus plugs.^{8,11,12} Computed tomography has been increasingly used in small animals to explore respiratory signs, especially when radiography reveals no specific abnormalities, as has been reported in up to 23% of cases of asthma and chronic bronchitis.^{4,13–15}

Bronchiolar plugging is well described in human medicine as a centrilobular (= anatomically located centrally within secondary pulmonary lobules) bronchiolar or small airway dilation, filled with mainly mucus, pus, fluid, or (rarely) tumor emboli.^{16–20} This produces on CT a

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characteristic pattern of a “tree-in-bud,” more commonly seen in lung periphery and associated with nodular and linear branching opacities, likened to a budding tree in spring.^{16,18–20} This specific pattern can be recognized when the nodules follow the airway tree, are evenly spaced from one another and do not contact the visceral pleural surfaces on CT.^{21,22} It is often associated with abnormalities of the larger airways (= bronchi) in humans.^{16,19} The authors have observed nodular and linear branching peripheral opacities on feline thoracic CT studies, similar to the description of the “tree-in-bud” pattern in humans. In clinical practice, the authors have also noted that the CT “tree-in-bud” pattern may appear as pulmonary nodular opacities on feline thoracic radiographs and mimic parenchymal nodules such as metastasis. Very little is available in the veterinary literature on the feline CT “tree-in-bud” pattern, even though it is very well described in humans and may represent an important pitfall in feline patients with or without respiratory symptoms.^{15–20,23}

Aims of the current study were to characterize the CT appearance of the “tree-in-bud” pattern in a group of cats, and compare the CT appearance of this pattern with thoracic radiographs and clinical findings.

2 | MATERIAL AND METHODS

The study was a retrospective, descriptive, multi-center design. The electronic medical record databases of the Centre Hospitalier Universitaire Vétérinaire d'Alfort, Centre Hospitalier Universitaire Vétérinaire de Montreal, Veterinary Medical Teaching Hospital of the University of Missouri and North Carolina State Veterinary Hospital were searched between January 2012 and September 2015, for thoracic CT reports containing descriptions consistent with a “tree-in-bud” pattern, using the key words “mucus,” “plug,” “plugging,” and “broncholithiasis.” Cats were included in the study if high-quality lateral and ventrodorsal or dorsoventral thoracic radiographic views were available, within a lapse of time of less than a month. Final decisions for subject inclusion or exclusion were made by one radiologist (P.P.).

Medical records for each included cat were recorded by one observer from each center and reviewed by two internists (G.B. and C.R.). Recorded data were as follows: breed, gender, neutered status, age, body weight, history, clinical signs, and time between the CT and the diagnosis. Presence of eosinophilia on complete blood count, findings on bronchoscopy, cytology (bronchoalveolar lavage, endotracheal wash, or fine-needle aspiration), percentage of eosinophils and neutrophils on bronchoalveolar lavage or endotracheal wash, presence of bacteria and parasites on bronchoalveolar lavage, treatment, and outcome, as well as histology or necropsy results, when available, were also recorded.

Digital images from all included thoracic CT and radiographic studies were retrieved and reviewed by two board-certified veterinary radiologists (S.S. and P.P.). All CT images were viewed using an offline workstation (Apple, iMac, Cork, Ireland; Osirix, v. 5.8.2., 64-bit, see www.osirix-viewer.com, Bernex, Switzerland) using a lung window (window width 1400; window level –550). A consensus on the imaging characteristics was reached for each cat. Radiologists used the

view they considered most appropriate for recording their findings, and multiplanar reconstruction and minimal intensity projections were used as needed for CT studies to assess the intraluminal location of bronchial plugs. The left cranial lobe was assessed as two different lung segments.

The following radiographic criteria were recorded for comparisons with the CT “tree-in-bud” pattern: affected pulmonary segment(s) (name and number), regional distribution in the segment (ventral-mid-dorsal), shape (nodular or tubular), number of lesions, maximal and minimal size, opacity (soft tissue or mineral), contours (well-defined or blurred). Globally, with respect to the lung and thorax, the following radiographic parameters were also recorded for each patient: abnormal lung pattern (including bronchial, peribronchial (= interstitial cuffing around a bronchial structure), unstructured interstitial, alveolar, nodular, or miliary), abnormal dilation of bronchial airways, parenchymal bands (defined as nontapering peripheral linear opacities extending from the visceral pleura), lung hyperinflation, pulmonary atelectasis, rib fractures, pneumomediastinum, and hiatal hernia.¹⁶ The diameter of pulmonary arteries was measured and compared at a given level from the heart to the accompanying vein, and with the 4th rib on lateral projections and the 9th rib on the dorsoventral or ventrodorsal projections.²⁴

The “tree-in-bud” pattern was evaluated on CT according to the following criteria: affected pulmonary segment(s) (name and number), regional distribution in the segment (ventral-mid-dorsal), distance between the lesion and the pleura (more or less than 5 mm), shape (nodular or tubular), number of nodular opacities, attenuation (Hounsfield units, HU), contours (well-defined or blurred), presence of broncholiths (= mineralized material within lumen of bronchial tree). Other qualitative assessments were subjective evaluation of bronchial wall thickening, bronchiectasis, peribronchovascular interstitial thickening, ground-glass opacity (defined as hazy increased attenuation that does not obscure visibility of underlying vasculature), parenchymal bands, mosaic attenuation pattern (defined as patchy regions of different lung attenuation), honeycombing pattern (defined as clustered cystic air spaces usually subpleural), lobar hyperinflation, pulmonary atelectasis, rib fractures, pneumomediastinum, hiatal hernia, and subjective tracheobronchial lymphadenopathy.¹⁶ Lobar hyperinflation was defined as a clear difference in attenuation and volume between two adjacent lobes, with the hyperinflated lobe presenting with a lower attenuation, larger lung volume, and convex borders with concomitant mediastinal shift away from the hyperinflated lobe. In order to quantify bronchiectasis, measurements of the bronchoarterial ratio were performed for each pulmonary segment, in each cat using the previously published method.²⁵ (The measurement was not attempted for atelectatic lobes.) Additionally, size of pulmonary arteries accompanying the bronchial tree plugs was measured: the pulmonary vessel diameter at the level of the “tree-in-bud” pattern was compared to the diameter of another lobar pulmonary vessel at a similar level of vascular ramification.

A consensus between one veterinary radiologist (P.P.) and two veterinary internists (C.R., G.B.) was reached to assign a final diagnosis category for each cat, based on review and discussion of all available historical, clinical information, and laboratory/imaging results. Cats were

TABLE 1 CT units and technical factors of parenchymal reconstructions in the four different centers

Hospital	CT units	KVp	mA	Matrix	Display field of view (cm)	Pitch
Alfort	Brilliance, 64-slice, Phillips, Amsterdam, Netherlands	120	196	768 × 768	35	0.5
Montreal	Light Speed, 16-slice, General Electric Healthcare, Medical Systems, Milwaukee, WI	120	250	512 × 512	10–12	1.0
Missouri	Aquilion, 64-slice, Toshiba America Medical Systems, Tustin, CA	120	250	512 × 512	10–12	1.0
NCSU	Sensation Somatom, 64-slice, Siemens, Malvern, PA	120	100–360	512 × 512	14.5–21	1.1–1.3

considered to have a definite diagnosis of bronchial disease if bronchoscopy and airway cytologic results were available and consistent with this diagnosis. Asthma was diagnosed if there was eosinophilic airway inflammation; chronic bronchitis if there was neutrophilic airway inflammation. Cats had a suspected diagnosis of bronchial disease if historical, clinical information, imaging results, and response to treatment were in favor but diagnostic tests (e.g., bronchoalveolar lavage fluid cytology) were missing.

3 | RESULTS

Thirty-six cats met the inclusion criteria. Mean age was 11.3 years (range: 8 months–20 years) and mean weight was 4.5 kg (range: 2.4–7 kg). Represented breeds included Domestic short, medium or long hair ($n = 24$), Siamese ($n = 6$), Ocicat, Persian, Burmese, Sphynx, Maine Coon, and Singapoura cats (one of each). There were 19 females and 17 males. Thirty cats had a history of one or more respiratory signs including cough (14); nasal discharge, sneeze, or stertor (13); tachypnea or increased effort of breathing (12); episodes of respiratory distress (7); and stridor (3). Six cats had no previous history of respiratory signs. Seventeen cats had abnormal lung sounds on auscultation including increased bronchovesicular sounds (13), wheeze (4), crackles (2), and diminished heart or lung sounds (1). Five cats presented with other systemic signs, including hyporexia or anorexia (3) and weight loss (2). Eight cats presented with a heart murmur or arrhythmia. A complete blood count was available in 32/36 cats and five presented with eosinophilia. Bronchoscopy was performed in 12/36 cats and results were available for ten. In all, an abnormal amount of mucus was observed leading to partial or complete obstruction of the airway. Ten cats had a bronchoalveolar lavage performed; five had eosinophilic inflammation, three had neutrophilic inflammation. An endotracheal (blinded) wash was performed in four; two cats had neutrophilic inflammation, one had a bacterial infection.

Thoracic radiographs were available for all 36 cats. Inspiratory right or left lateral and ventrodorsal or dorsoventral projections of the thorax were performed using digital radiographic systems. Flat panels (Canon CXDI 50 G flat panel digital systems, Marne-la Vallée, France;

TABLE 2 Computed tomographic and radiographic most frequent affected pulmonary segments, thoracic distribution, and shape of tree-in-bud lesions

	RX: number of cats	CT: number of cats
Most frequent affected pulmonary segment	R cr: 13 R cd: 12 L cd: 12	R cr: 27 R cd: 23 Acc: 18 Cr of L cr: 18
Thoracic distribution of lesions	Ventral: 17 Dorsal: 14 Middle: 10	Ventral: 33 Dorsal: 24 Middle: 21
Lesion shape	Nodular: 22 Tubular: 7 Both: 1 No lesion: 6	Nodular: 5 Tubular: 20 Both: 11
Lesion size (mm)	Min: 1 mm, Max: 9 mm	Min: 0.7 mm, Max: 9 mm

R cr, right cranial; R cd, right caudal; L cd, left caudal; Acc, accessory; Cr of L cr, cranial segment of left cranial. Note that patients can have more than one pulmonary segment affected and more than one lesion location. Total number of patients = 36.

Uxbridge, G-B; Melville, NY) were used in two institutions. A computed radiography system (Siemens Iconos, Oakville, ON, Canada) was used in one institution and a charge coupled device-based digital radiography system (IDC Imaging Dynamics Corporation, Calgary, AB, Canada) in the last center. In six cats, no “tree-in-bud” pattern was radiographically visible, despite its presence on CT; none of these cats were asymptomatic. All lesions identified on radiographs were confirmed on CT as having a bronchial tree location. The number of pulmonary segments affected by the “tree-in-bud” pattern on radiography was as follows: 1 ($n = 17$), 2 ($n = 8$), 3 ($n = 2$), 4 ($n = 1$), or 7 ($n = 2$) segments. The most frequently affected segments are found in Table 2. The thoracic regional distribution of the “tree-in-bud” pattern was either ventral in 17 cats, dorsal in 14 or in the middle of the lobe in 10 (Figures 1 and 2) (Table 2). Lesions were nodular in shape ($n = 22$) (Figure 1), tubular ($n = 7$) or both ($n = 1$); they ranged from 1 to 9 mm (Table 2) and quantitatively, from 1 ($n = 2$) to innumerable ($n = 6$). They were mineralized in two cats and of soft-tissue opacity in the remainder. Their contours were either well defined (77%) or blurred (23%). The size of the adjacent pulmonary artery was within the normal range in all cats (no larger than the diameter of the fourth (cranial) or ninth rib

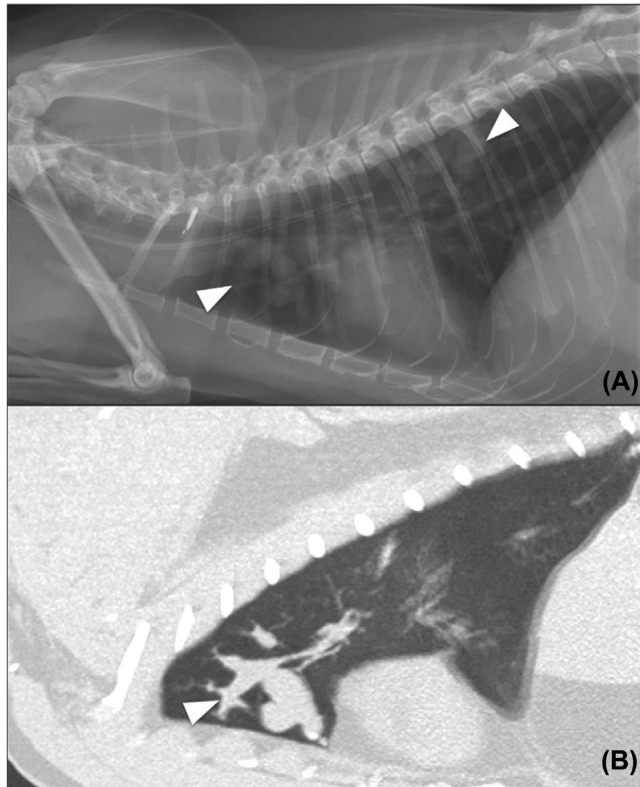


FIGURE 1 A, Lateral radiograph of the thorax of an 8-year-old mixed Siamese cat, presented for acute vomiting. No respiratory signs were reported. Thoracic radiographs were taken to check the position of the nasoesophageal tube. Note the presence of several well-defined nodular soft tissue opacities (arrowhead) grouped along the axis of the bronchial tree. A diffuse bronchial pattern was also observed. A computed tomographic exam of the thorax was performed to assist for differentiation between a “tree-in-bud” pattern and metastases. B, Multiplanar reconstruction in minimum intensity projection (0.5 mm thick slice) acquired with high resolution algorithm displayed in lung window (WL: -500; WW: 1400) showing the intraluminal location of the lesion with a tubular shape (arrowhead) (vs. nodular on radiograph), creating a distortion of the bronchus. The exam was performed on the awake patient, hence the motion artifact on the caudal lung

(caudal arteries)). Dilation of airways (evaluated at a different location from the airways with plugs) was observed in nine cats. All cats but two had an abnormal lung pattern: bronchial ($n = 34$), peribronchial ($n = 4$), alveolar ($n = 10$), unstructured interstitial ($n = 2$), or miliary interstitial ($n = 1$). Nine cats presented with lobar atelectasis of one or two pulmonary segments: the right middle lobe ($n = 2$), right cranial ($n = 3$), and cranial ($n = 4$) or caudal ($n = 2$) segment of the left cranial lobe were affected. Other lesions noted included lung hyperinflation (16), parenchymal bands (6), rib fractures (5), and hiatal hernia (2). No tracheobronchial lymphadenopathy or pneumomediastinum was observed.

Thoracic CT scans were obtained using four different CT units in the four different centers (Table 1). Examinations were performed under general (inhalational or constant rate infusion intravenous) anesthesia, awake or under light sedation in a dedicated patented device (Universal Medical Systems, Inc., Cleveland, OH) or similar customized plexiglas-made system. Inspiratory breath hold CTs were performed

by manually inflating the lungs with a long Bain breathing system and blocking expiration during CT acquisition in two centers (not performed in the other universities). Expiratory CTs were acquired in one center by giving a bolus of propofol (PropoFlo (10 mg/mL), Abbott Laboratories, North Chicago, Ill) and disconnecting the patient from the ventilator during the acquisition. A critical care ventilator (Engstrom Carestation, GE Healthcare, Fairfield, CT) was used to perform inspiratory and expiratory breathholds in the other center, according to the previously described technique.²⁶ All thoracic CT images were acquired in sternal recumbency. Images were reconstructed with a high-resolution algorithm and 1-mm thick sections, no interval.

Eighteen cats had paired inspiratory/expiratory CT performed; six cats had only expiratory CT; the other 12 had undifferentiated respiratory-phased CT. The number of pulmonary segments affected by the “tree-in-bud” pattern was recorded for each cat: 1 ($n = 6$), 2 ($n = 7$), 3 ($n = 6$), 4 ($n = 6$), 5 ($n = 2$), 6 ($n = 5$), or 7 ($n = 4$) segments were affected. Among the asymptomatic cats, 1 ($n = 1$), 2 ($n = 2$), 3 ($n = 1$), 4 ($n = 1$), or 5 ($n = 1$) segments were affected. The most frequently affected segments are found in Table 2. The regional distribution of the “tree-in-bud” pattern was ventral in 33 cats, dorsal in 24 and in the middle of the lobe in 21 (Table 2). It was always observed in the airway tree at a distance greater than 5 mm away from the closest pleura. The shape of the lesions was nodular ($n = 5$), tubular ($n = 20$) or both ($n = 11$) (Figure 1). When tubular, the lesions appeared as V or Y branching lines. When nodular, the number was counted and ranged from one ($n = 2$) to innumerable ($n = 8$); their size ranged from 0.7 to 9 mm (Table 2). Their attenuation ranged from 10 to 2355 HU, with a median of minimum values of 53 HU and a median of maximal ones of 195 HU. None of these was enhancing after intravenous injection of contrast media. Ten cats had small mineralized opacities within the airways, consistent with broncholiths (Figure 3). The contours of the lesions were either well defined (86%) or blurred (14%). When more than one focal lesion was observed, a characteristic pattern imitating a “tree-in-bud” was always observed, with nodular and linear soft-tissue or mineral branching opacities, in the airway tree (Figure 2). The size of the adjacent pulmonary artery was normal ($n = 25$), decreased ($n = 10$) (Figure 4) or enlarged ($n = 1$). An abnormal CT pattern was observed in all cats (some with more than one abnormality) and consisted of thickened bronchial walls ($n = 35$), parenchymal bands ($n = 25$) (Figure 4), mosaic pattern ($n = 12$) (Figure 5), ground glass opacity ($n = 10$), honeycombing ($n = 10$) (Figure 6), or increased opacity of the peribronchovascular interstitium ($n = 8$). Lobar hyperinflation was assessed as present in 11 cats on CT (Figure 7). The six asymptomatic cats had the following pulmonary lesions: bronchial thickening (6), lobar emphysema (2), ground glass opacity (2), plate-like atelectasis (2), mosaic attenuation (1), honeycombing (1), peribronchovascular thickening (1). The broncho-arterial ratio measured to objectively assess bronchial dilation showed evidence of bronchiectasis in all cats, in at least one pulmonary segment (88% cats in the cranial segment, 87% in caudal segment of left cranial, 82% in left caudal, 97% in right cranial, 84% in right middle, 83% in right caudal, 66% in accessory). Nineteen cats had evidence of lobar atelectasis in 1 ($n = 11$), 2 ($n = 4$), 3 ($n = 3$), or 4 ($n = 1$) segments. The right middle lobe was most commonly affected (47% cases); the right caudal lobe was the only lobe never

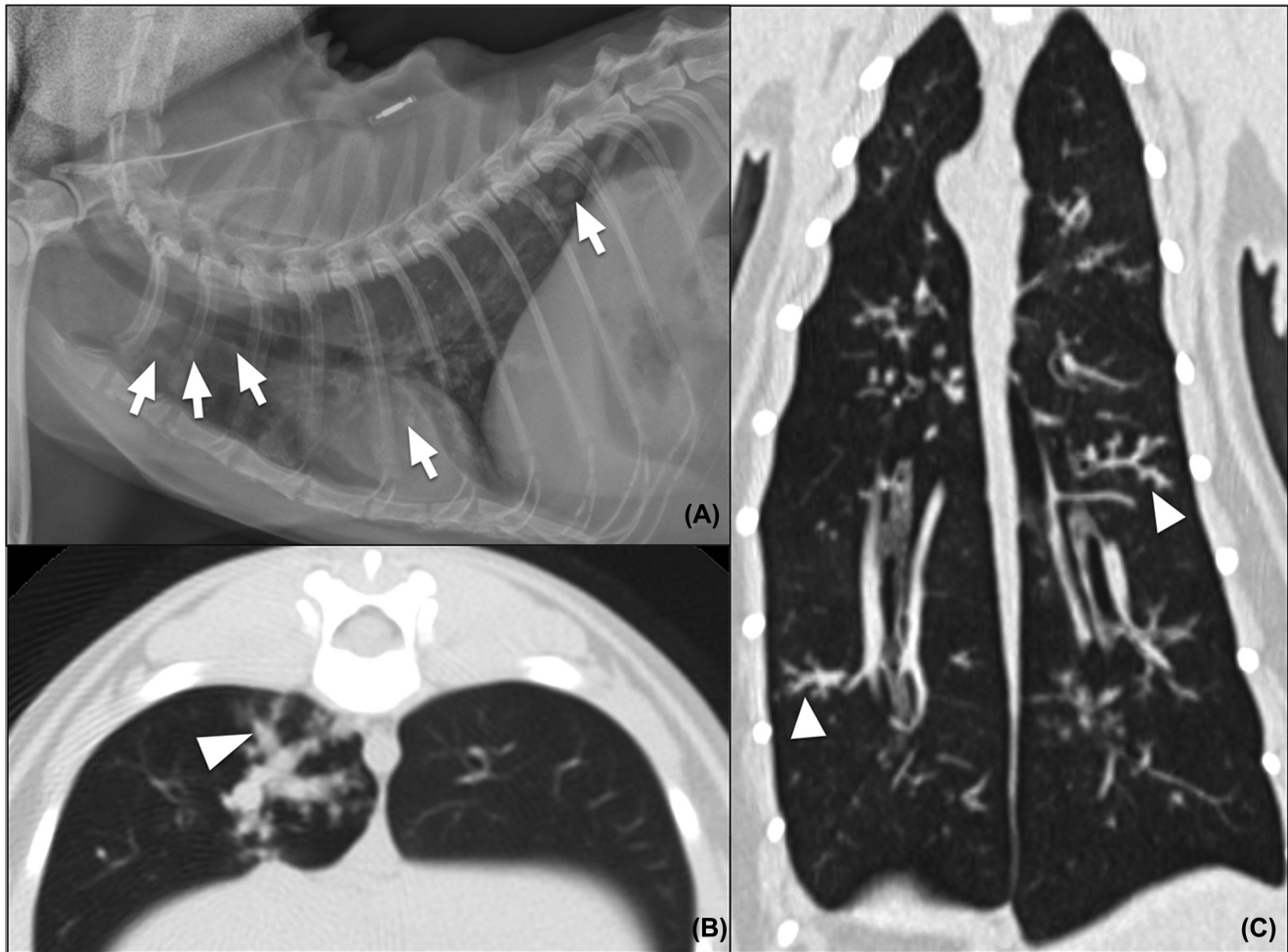


FIGURE 2 Lateral radiograph of the thorax of a 16-year-old Domestic shorthair cat, presented for evaluation of a 9-month history of nasal congestion and clear-mucoid nasal discharge. The patient sneezed several times daily. A, Three small soft tissue nodular opacities are visible in the cranioventral lung field (arrow). Computed tomographic images of the same patient illustrating several areas of “tree-in-bud” pattern (arrowhead) on transverse (B) and dorsal (C) multiplanar reconstruction, displayed in lung window (WL: -500; WW: 1400). The “tree-in-bud” pattern was more commonly seen in lung periphery (arrowhead) and associated with nodular and linear branching opacities likened to a budding tree in spring

affected. Other lesions included rib fractures (5) and hiatal hernia (2); tracheobronchial lymphadenopathy or pneumomediastinum were not observed in any cat.

Results of the multidisciplinary consensus were that 12 cats had a definite diagnosis of feline bronchial disease: five cats had asthma, five had chronic bronchitis, one had a bacterial bronchial infection, one had an unknown disease. Regarding the other 24 cats, 23 cats were suspected to have a bronchial disease, but lacked airway cytologic results to confirm it. Three of these cats had concurrent pulmonary neoplasia (two malignant epithelial tumors, one fibrosarcoma metastasis). One cat had an unknown disease.

4 | DISCUSSION

To the authors' knowledge, this is the first study focused on describing the CT and radiographic features of a feline “tree-in-bud” pattern. On CT, the “tree-in-bud” pattern demonstrated a characteristic appear-

ance of nonenhancing nodules and branching V or Y lines, representing enlarged bronchial airways coursing perpendicular and parallel to the CT plane section. As the majority of cats with a “tree-in-bud” pattern in this study were diagnosed or suspected to have an underlying feline bronchial disease, we suggest that the “tree-in-bud” pattern represents an imaging feature of this disease. Based on the bronchial localization and the presence (or suspicion) of lower airway disease, we also presume the “tree-in-bud” pattern represents bronchial tree plugging, as described in humans.

The “tree-in-bud” pattern affected cats of a wide spectrum of age and breed, with no apparent gender predisposition. Coughing, either alone or with other respiratory signs, was the most common complaint among the population (14/36 cats). However, a “tree-in-bud” pattern can also be found in the absence of respiratory signs, as this was the case for six cats, which had mild to severe “tree-in-bud” pattern, associated with pulmonary changes. Thoracic lesions, especially bronchial disease and mucus plugging in asymptomatic cats have previously been reported.¹⁵ Thus, a “tree-in-bud” pattern can also be subclinical.

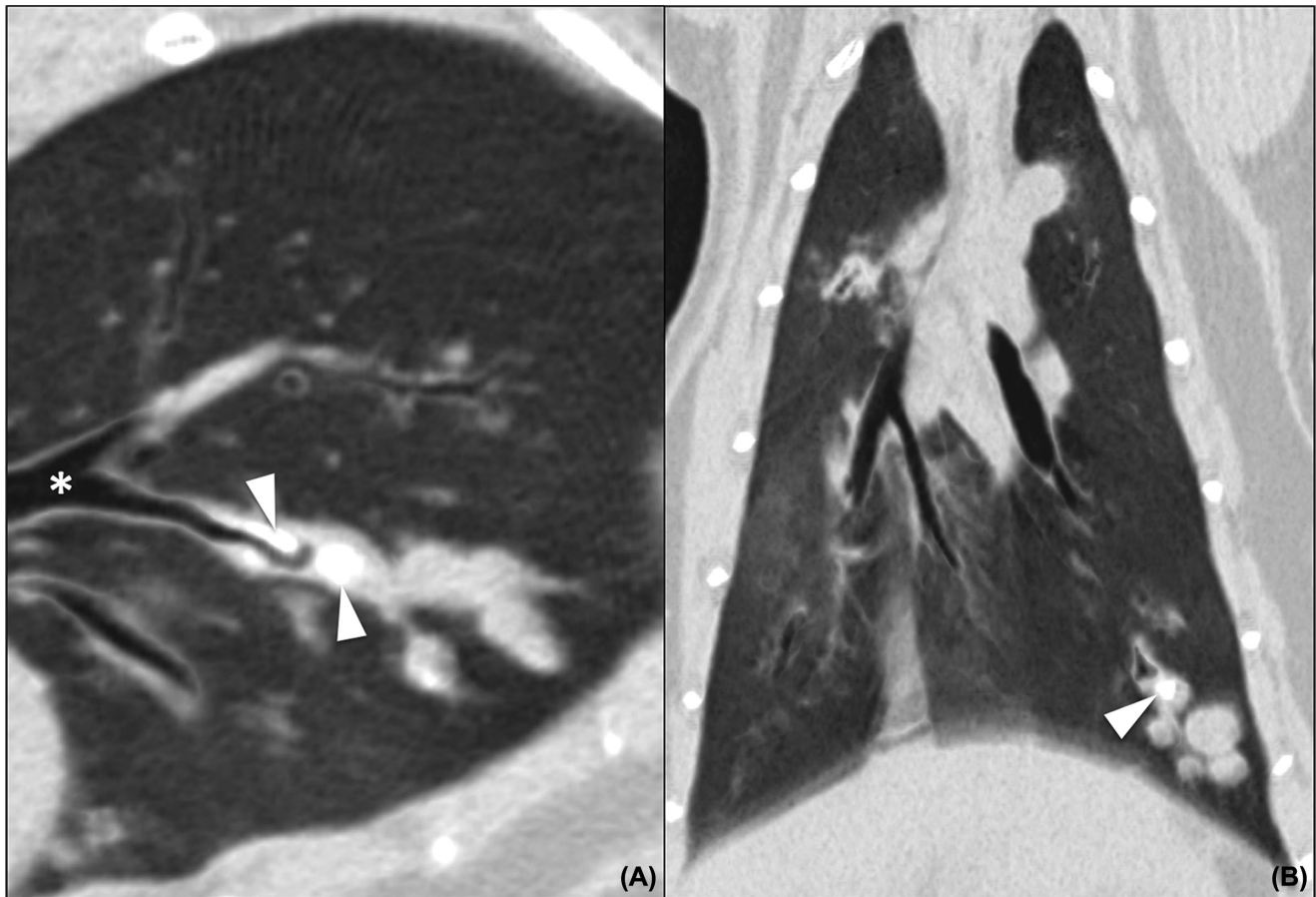


FIGURE 3 Sagittal (A) and dorsal (B) multiplanar reconstruction of computed tomographic images, in minimum intensity projection (0.5 mm thick slice) and displayed in lung window (WL: -500; WW: 1400). Arrowheads are depicting endobronchial mineral densities, presumed as broncholithiasis in the lumen of the segmental bronchus (asterisk)

Another interesting finding is that 36% of cats had a history or clinical signs of rhinitis. Cats could have had two unrelated diseases leading to rhinitis and lower airway disease. Another possibility is the “united airways theory” which in humans and an experimental model of feline allergic asthma suggests an immunologic link to explain the comorbid conditions of rhinitis and allergic asthma.^{27,28} A third possibility is postnasal drip leading to chronic aspiration and chronic bronchitis with potential bronchial plugging.

In humans, on histologic examination, the “tree-in-bud” pattern reflects the impaction of the centrilobular bronchioles with mucus, pus, granulation tissue, or fluid, resulting in dilation of these small airways that should not normally be seen on CT or radiography.^{18,21} Cytology of bronchoalveolar lavage and histology of lung tissue were inconsistently performed in this retrospective study; thus the nature of these plugs in cats remains inconclusive. In humans, the differential diagnoses for “tree-in-bud” lesions include infectious agents (bacterial, viral, and fungal agents, tuberculosis), aspiration, infectious variants (cystic fibrosis, allergic bronchopulmonary disease), and noninfectious causes (bronchioalveolar carcinoma, follicular bronchiolitis, intravascular metastases).^{19,21,22}

In a large number of our patients, some “tree-in-bud” lesions were considered to be in a bronchiolar location on CT (defining bronchioles as airways lacking a cartilaginous wall and measuring < 2 mm).

Authors propose this could represent extension of the bronchial disease into the small airways. Further studies would however be necessary to assess the possible histologic extension of the inflammation from the large airways (or bronchi) to the small airways (or bronchioles).

Radiographs are considered to be of low utility to detect bronchiolar disease in humans.²⁹ In the current study, radiography led to an underestimation of the number of “tree-in-bud” lesions in 24/36 cats; moreover, its presence was completely missed in 6/36 animals. On radiography, the computed tomographic “tree-in-bud” pattern presented nodular shapes in 76% of cats. Radiography thus tended to artifactually increase the nodular appearance of the lesions, due to the two-dimensional superimposition, and potentially causing confusion with other types of pulmonary nodules such as metastasis, granulomas, or nodular fibrosis. Moreover, two cats only had a unique intraluminal lesion on CT and this could make the confusion with an interstitial nodule on radiographs even greater, since the characteristic pattern evoking a budding tree was less conspicuous. Thus, the computed tomographic “tree-in-bud” pattern may represent a pitfall on thoracic radiographs. However, the presence of multiple soft tissue nodules on radiographs limited to a single pulmonary lobe should raise the suspicion of “tree-in-bud” pattern, as parenchymal metastatic disease usually tends to affect multiple lung lobes.

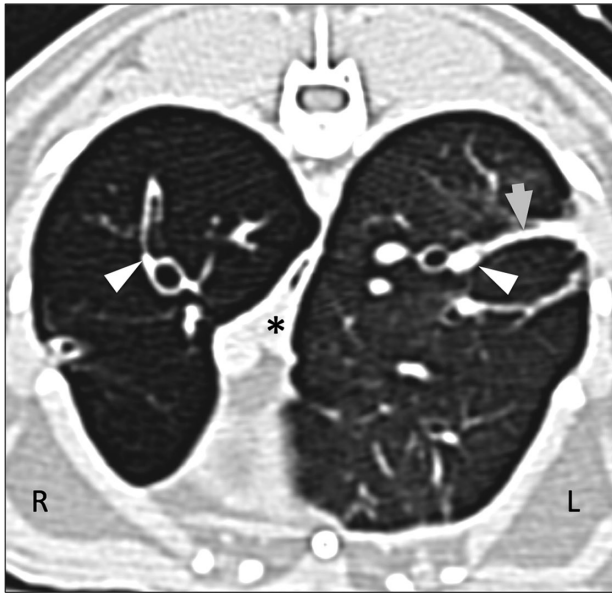


FIGURE 4 Transverse computed tomographic image of the thorax displayed in lung window (WL: -500; WW: 1400). A decreased size of the pulmonary artery is observed in the right (R) caudal lung lobe (arrowhead), compared to the pulmonary artery depicted by the other arrowhead in the left (L) caudal lung lobe. Note the subjective difference in attenuation between the left and right caudal lung lobes. A right-sided mediastinal shift is observed but was considered secondary to atelectasis (asterisk) of the accessory lobe. Note also the presence of a parenchymal band (gray arrow) defined as non-tapering peripheral linear opacities extending from the pleura

On CT, no enhancement of the “tree-in-bud” pattern was observed after IV contrast media injection; this can also be a helpful tool in some cases to differentiate pulmonary nodules associated to the bronchi and bronchial tree plugs.

A bronchial pattern was observed in all cats on radiography; on CT, thickening of bronchial walls was almost always observed (97%). There was thus good agreement between radiography and CT regarding the evaluation of bronchial walls. In this study, the majority of cats had bronchial disease justifying the presence of a bronchial pattern in both modalities. However, the distinction on radiographs between bronchial and peribronchial lesions can be challenging given the similarity in radiographic aspect and could have led to overestimation on radiographs. Honeycombing with a subpleural predominance was observed in 10 cats (28%). Honeycombing in people represents destroyed and fibrotic lung tissue containing numerous cystic airspaces with thick fibrous walls, observed in the late stage of various lung diseases, with complete loss of acinar architecture.^{16,22} It is a CT feature of established pulmonary fibrosis in people.¹⁶ Thus, these animals most likely had some degree of pulmonary fibrosis.

On radiography, all adjacent vessels appeared normal in size, whereas on CT 28% cats had a decrease in size of the accompanying pulmonary artery. In humans, bronchiolar obstruction leads to downstream hypoventilation of the alveoli, hypoxic vasoconstriction, and a decrease in size of affected vessels; simultaneously blood is shunted to other better oxygenated regions of the lung, leading to a pattern of mosaic attenuation.^{17,21} Thirty-three percent of cats had a

mosaic pattern on CT, associated with the “tree-in-bud” pattern. Pulmonary hyperinflation was visible on radiography in 44% cats, but in only 31% of cats on CT. It is challenging to compare the two modalities with respect to pulmonary hyperinflation as radiography evaluated global lung volume in awake and spontaneously breathing cats (ideally obtained at the end of inspiration), whereas CT scans were acquired under more variable circumstances including awake, sedated, and anesthetized cats. Additionally, to our knowledge, no absolute pulmonary attenuation cut-off value exists to diagnose hyperinflation in cats. Inspiratory breath-hold anesthetized CT scans have previously been shown to facilitate discrimination between asthmatic and healthy cats in comparison to unrestrained CT scans.²⁶ Mosaic attenuation caused by regions of air trapping can only be definitively detected and discriminated from mosaic attenuation caused by vascular disease on CT using paired inspiratory and expiratory breathhold scans. Regions of the lung with air trapping show an accentuated pattern of mosaic attenuation on the expiratory series. This is due to air remaining in regions downstream of obstructed airways (i.e. decreased attenuation), showing a sharper contrast to adjacent regions with increased attenuation due to loss of air from exhalation. The major differential diagnoses for hyperinflation includes air-trapping secondary to a bronchial disease and pulmonary emphysema. Pulmonary emphysema, defined as a permanent and abnormal enlargement of air spaces distal to the terminal bronchioles and accompanied by destruction of the walls of the involved air spaces, has distinct CT features reflecting loss of normal tissue.

Parenchymal bands, bronchiectasis, and atelectasis were more often diagnosed on CT than radiography. Parenchymal bands were frequently observed on CT (69% cats) and underestimated on radiography (17% cats). As mentioned previously, this is likely a limitation of the two-dimensional nature of radiography and the fact that these bands are perpendicular to the pleura, likely being visible on only one radiographic projection. Bronchiectasis has previously been described in cats.^{30–32} According to the broncho-arterial ratio, all cats had bronchiectasis on CT. It is possible that this ratio was inappropriate to discriminate bronchiectasis with a good level of sensitivity and specificity since the diameter of the adjacent pulmonary artery was decreased in 28% of the cats in our study, which makes the pulmonary artery a variable reference. However, in humans, the “tree-in-bud” pattern is reported to be often associated with bronchiectasis.¹⁹ Lung atelectasis was observed in 53% of cats on CT and 25% of cats on radiography, most commonly affecting the right middle lung lobe and the cranial part of the left lung lobe, as found in previous studies.¹⁵ Anesthesia is most likely responsible for this finding, given the fact that most cats were awake during radiographs and anesthetized during CT, as previously mentioned.³³ However, all cats except one had radiographs performed before CT, with delays between the two exams extending up to 3–4 weeks for 11 cats, so disease progression (pneumonia for example) between the time of radiographs and CT can not be excluded. This large lapse of time for a few cats represents a limit to this study. The authors felt however that changes regarding the “tree-in-bud” lesions would not be major, given their chronic evolution, and that this lapse of time was acceptable in only a small number of cats.



FIGURE 5 Computed tomographic images illustrating a mosaic attenuation pattern, on transverse images (A) and dorsal multiplanar reconstruction (B) displayed in lung window (WL: -500; WW: 1400). There is a patchwork of regions of differing attenuation (black arrowhead). Note the presence of “tree-in-bud” pattern (gray arrow) (A) and atelectasis of the right middle lung lobe (asterisk) (B)

Radiography poorly identified intraluminal mineralization. “Tree-in-bud” pattern was mineralized in 6% of the cats on radiography and 28% of cats on CT. Computed tomography has a higher contrast-resolution than radiography, explaining the difference in results. In human medicine, the term broncholithiasis refers to mineralized elements originating outside the airway lumen migrating inside of and plugging the lumen; it differs from endobronchial material, which subsequently becomes mineralized. It is not possible to make this distinction in cats based on our study design and thus the term broncholithiasis, a differential for pulmonary mineralization, has been previously used.^{12,34} Their pathogenesis is not well understood but is speculated to result from dystrophic mineralization of an accumulation of bronchial secretions secondary to a chronic inflammatory or infectious pulmonary disease.¹² Broncholithiasis is also described in humans most commonly due to erosion by and migration of a calcified adjacent lymph node into the bronchial lumen, usually associated with tuberculosis or histoplasmosis.³⁵ Other causes include aspiration of bone tissue or in situ calcification of aspirated foreign material, and

erosion by and extrusion of calcified or ossified bronchial cartilage plates.

There are a few other limitations to this study. Prior to the interpretation of images, the radiologists were not blinded to the fact that radiographs and CT were from cats that already had a diagnosis of “tree-in-bud” pattern established. They were however blinded to the clinicopathologic data, the results of the other imaging modality and the final diagnosis. The multicentric and retrospective nature of this study led to different clinicians performing the clinical exams, different CT and radiographic protocols, and different anesthetic protocols. Finally, bronchoalveolar lavage cytology was inconsistently performed in patients: some were asymptomatic so there was no ethical nor clinical justification for general anesthesia and bronchoscopy; some owners declined further explorations and chose probabilistic treatment. Thus, a medical final diagnosis was not achieved in all cases.

This is the first detailed description of CT “tree-in-bud” pattern and radiographic correlates in cats. Feline “tree-in-bud” pattern has

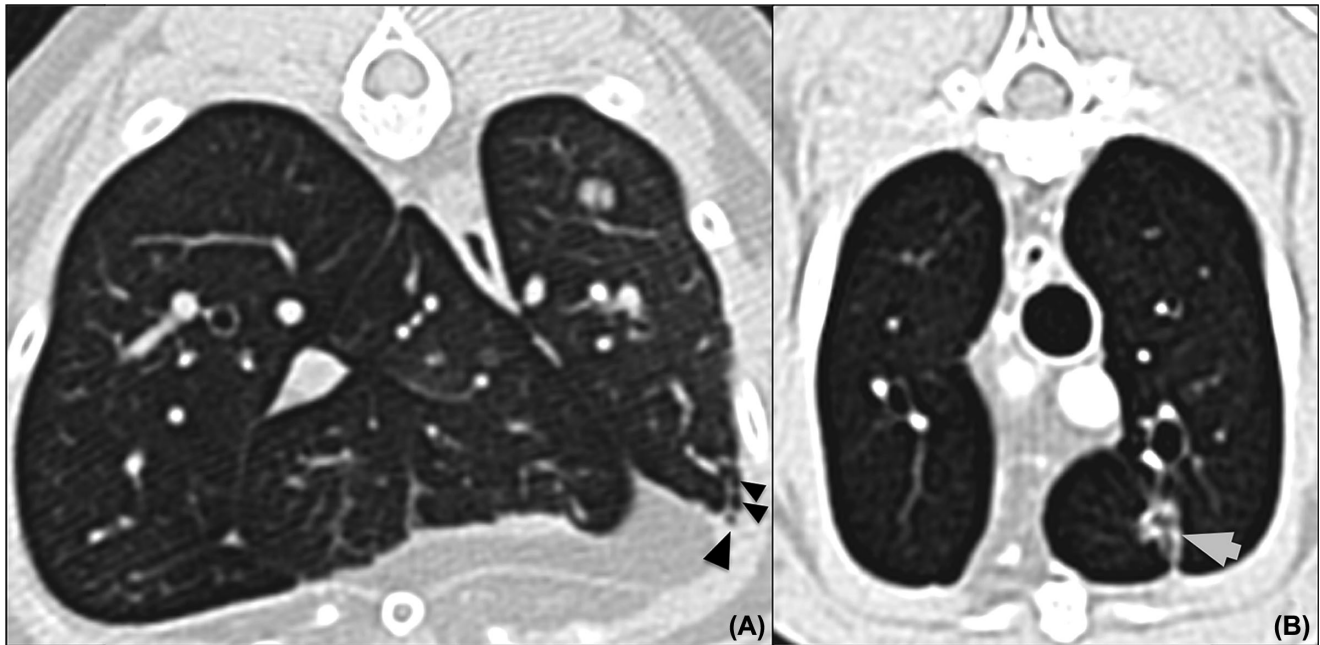


FIGURE 6 Transverse computed tomographic images displayed in lung window (WL: -500; WW: 1400), illustrating a honeycombing pattern (A) with clustered cystic air spaces in subpleural and ventral distribution (arrowhead) and dilation of the peripheral lower airways (gray arrow) (B)

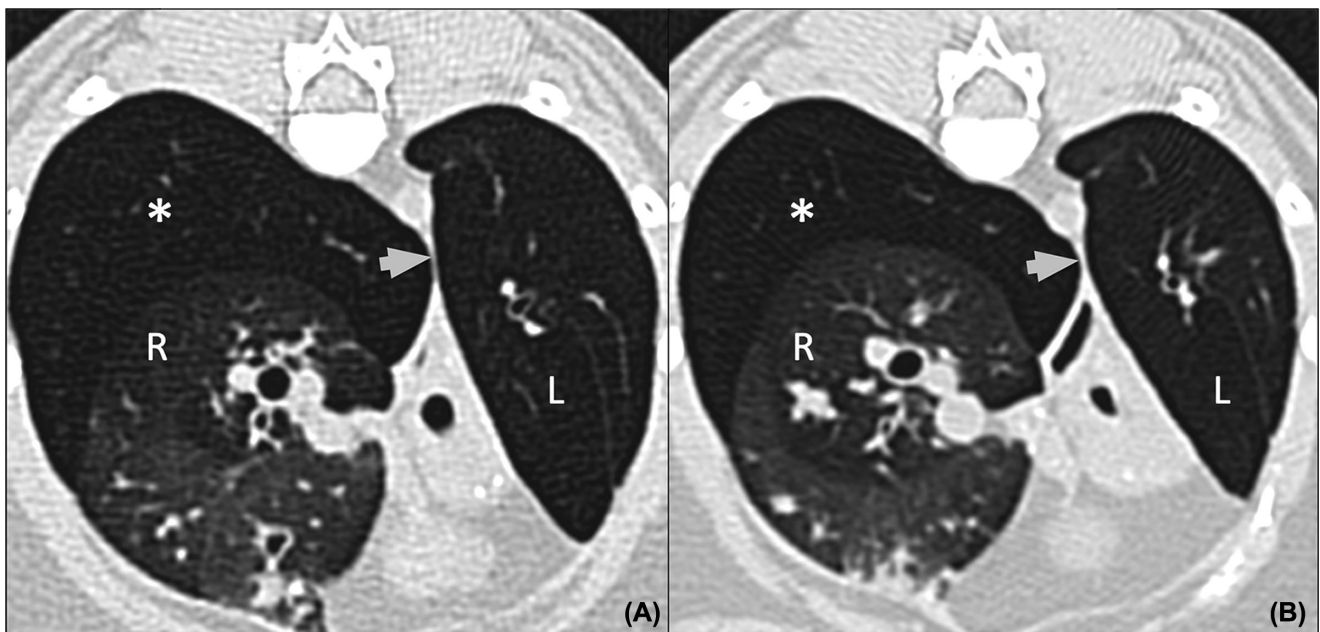


FIGURE 7 Transverse computed tomographic images of the thorax of a cat performed with inspiration (A) and expiration (B) breath holds technique displayed in lung window (WL: -500; WW: 1400). Note the decreased volume of the left caudal lung lobe (L) associated with an ipsilateral mediastinal shift (gray arrow). Note also that the right cranial lung lobe (asterisk) is visible dorsally to the right caudal lung lobe (R) in this caudal portion of the thorax, attesting its increased volume. Finally, there is a clear difference in attenuation between the right caudal (R) and the right cranial (asterisk) lung lobes that persists in expiration, suggesting air-trapping and lobar hyperinflation

similar characteristics to humans, leading to linear and nodular branching soft-tissue or mineral opacities on both imaging modalities. On radiographs, the number of “tree-in-bud” patterns and affected segments were often underestimated; furthermore, they may appear as soft-tissue peripheral nodules and should not be confused with parenchymal nodules such as metastatic disease. CT should be consid-

ered as a further diagnostic tool in feline patients with pulmonary nodules on radiographs to help distinguish between a “tree-in-bud” pattern and true parenchymal nodules or in feline patients with clinical signs of lower airway disease and normal thoracic radiographs. Moreover, the presence of a “tree-in-bud” pattern should raise the suspicion for an underlying feline bronchial disease.

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